Project goals: Small, repeatedly-flooded municipalities with older building stock have no engineering tool to evaluate whether community-scale drainage investment or distributed building hardening better reduces flood damage. Here we propose a closed-loop cyber-physical system (CPS) that couples the physical behavior of pre-code load-bearing masonry and timber, absent from existing fragility frameworks, with a computational core that turns free national data and community memory into building-level damage estimates. It senses through USGS gauges and NOAA precipitation operationally, with commercial ICEYE Synthetic Aperture Radar (SAR) for one-time building-level duration and depth calibration and open Sentinel-1 SAR for flood-extent context; its computational core is a Bayesian uncertainty-quantification framework fusing a GIS-parameterizable mass balance model, one whose inputs come entirely from free national GIS data layers (USGS 3DEP, NLCD, SSURGO, NHD StreamStats) rather than locally calibrated hydraulic models, with community-elicited duration priors and a gauge-stage-to-terrain projection for building-level depth; and the community closes the loop through a co-developed two-mode interface for scenario-based retrofit planning, forecast-driven staging, and post-event model updating. **RQ1** (Napolitano, Wauthier, and Busse) asks to what extent this CPS, operating on free national data and community knowledge, can produce building-level damage assessments decision-adequate to weigh community-scale drainage against building-scale hardening on a common probabilistic basis, without hydraulic simulation. **RQ2** (Napolitano with input from all PIs) asks under what conditions locally held community knowledge, entered as Bayesian calibration priors, improves damage assessments relative to a GIS-only baseline, and where it functions as a substitute versus complement to geospatial data. The intellectual contribution is the closed loop these questions form, in which free national data and community knowledge together support real investment decisions, with community knowledge a structural calibration input measured against a geospatial baseline, the Community-Inspired Research (CIR) component of this CPS.

1 Status quo: Small communities at the confluence of riverine flood hazard and older building stock face a resilience challenge that no existing engineering framework is designed to address. Across the northeastern United States and Appalachian corridor, thousands of towns share this profile of multi-day riverine flood exposure, older load-bearing masonry and timber cores, and institutional capacity below the threshold for hydrodynamic simulation [1, 2]. Of the more than 22,000 communities FEMA's National Flood Insurance Program enrolls nationwide [3], thousands in this region combine significant pre-1940 building stock, no in-house hydraulic modeling capacity, and repeated riverine flooding. In Vermont alone, over a quarter of the housing stock predates 1939 [4, 5]. For these communities, every flood event surfaces the same unresolved question. Where do limited flood-mitigation dollars belong?

These communities face a menu of mitigation choices but lack any way to weigh them on a common probabilistic footing using the assets they actually have: a handful of physical sensors, public geospatial data, and community memory. The sharpest case is the allocation question: would a community-scale drainage investment reduce damage enough to justify its cost over the same dollars spent hardening individual buildings? Answering it means linking upstream drainage decisions to downstream building damage probabilities, a comparison no existing framework lets a small community make.

That comparison demands two things these communities do not have. First, a model: hydrodynamic simulation pipelines such as HEC-RAS and LISFLOOD-FP require calibrated hydraulic models and sustained technical staff that small towns cannot maintain [6–8]. Second, data on how their own buildings fail: damage functions in HAZUS and FEMA P-58 are built around post-1945 construction, so the load-bearing masonry and timber structures built before modern building codes (pre-code), the dominant type in repeatedly flooded historic downtowns, are absent from existing fragility frameworks [9, 10]. Those frameworks also treat damage as a function of depth alone, not the extended-inundation degradation that governs older masonry. Even NIST SP 1310 [11] concedes that its framework is too complex for smaller agencies and calls for simplifications that smaller organizations can implement [11, Sec. 6.1.3].

What these towns lack in models and data, they hold in memory: how deep the water rose, how long it stayed, and which buildings took the worst of it. Existing community resilience frameworks are largely top-down, and none gives a community a way to put this knowledge into a probabilistic assessment. We treat that knowledge not as context but as a calibration input, locally held flood memory entered as Bayesian priors, whose contribution can be measured directly against a geospatial baseline. Building that measurement into a working loop is the work that follows.

1.1 Research questions and intellectual merit:

- **RQ1 (Napolitano, Wauthier, and Busse):** To what extent can a CPS operating on free national data and community knowledge, without hydraulic simulation, produce building-level damage assessments decision-adequate to weigh community-scale drainage against building-scale hardening on a common probabilistic basis?
- **RQ2 (Napolitano with input from all PIs):** Under what conditions does locally held community knowledge, entered as Bayesian calibration priors, improve flood damage assessment accuracy and calibration relative to a GIS-only baseline, and where does it function as a substitute versus complement to geospatial data?

The central contribution is the closed loop these questions form, a working CPS in which free national data and community memory together produce building-level damage assessments decision-adequate for real investment choices, with no hydraulic simulation. Realizing it required three advances. The first is **computational**: a Bayesian framework that infers building-level depth and duration from sparse, interval-censored observations, fuses them with a GIS-parameterizable mass balance model and community-elicited priors, and propagates uncertainty to a decision-adequate damage estimate. The second is **physical**: the CPS senses flood signatures and informs physical interventions (drainage retrofits, building hardening) whose effects reenter future flood behavior. The third treats **community knowledge** as a structural calibration input, characterizing when Bayesian priors from local knowledge improve accuracy over a GIS-only baseline and where they function as substitutes versus complements. Together these establish that free national data fused with community knowledge suffices for the investment decisions small municipalities face.

2 Current state-of-the-art: Closing the loop requires four capabilities: estimating how long water stays at building level, translating duration into structural damage, calibrating with community knowledge where data are sparse, and converting damage probabilities into investment decisions whose outcomes feed back into the next event. Each has a mature literature, yet none addresses the interfaces among all four. The broader CPS literature does not close it either: community sensing architectures for flood monitoring assume local sensor infrastructure or sustained community effort that resource-constrained municipalities cannot maintain, and none carries the community knowledge mechanism, independent ground truth, or physical damage model the loop requires.

2.1 Hydrological mass balance modeling and flood duration estimation: Hydrological mass balance models estimate flood volume, storage, and drainage timescale from watershed geometry, land cover, soil properties, and precipitation. Unit hydrograph and rational method variants provide event-scale peak discharge for small watersheds [12]; GIS-parameterized implementations using USGS StreamStats, NHD Plus, NLCD, and SSURGO extend these to ungauged basins [13, 14]. Prior work has applied these methods to peak discharge and flood extent at community scale; their application to building-level inundation duration estimation has not been characterized. Whether a GIS-only mass balance model can produce building-scale drainage timescale estimates accurate enough for retrofit allocation is unknown (Subtask 1.2).

2.2 Fragility and vulnerability frameworks for flood damage: Flood fragility assessment relies on depth-damage functions mapping peak depth to expected damage ratio [9, 15, 16]. These share a structural assumption: peak depth is the governing input, and damage accumulates instantaneously. For post-1945 construction that is approximately correct; for pre-code load-bearing masonry it is not. Extended inundation drives mortar softening, progressive brick saturation, and rising damp that depth-damage functions cannot capture [17–19]. European riverine flood studies document that masonry damage state is better predicted by depth and duration together than by depth alone [20–22], yet no duration-dependent fragility surface exists for the pre-code masonry archetypes that dominate repeatedly flooded northeastern American historic cores. This absence blocks CPS integration: a duration-producing computational core needs duration-dependent fragility, which no existing framework provides for this stock.

2.3 Community knowledge in probabilistic risk assessment: Community knowledge constitutes a substantial information resource that formal modeling pipelines largely bypass. The Sendai Framework prioritizes community-based approaches [23]; CBDRR programs have shown that locally held knowledge identifies hazard exposures and infrastructure conditions that national datasets miss [24], and participatory GIS methods have supplemented remotely sensed flood observations [25, 26]. Existing approaches treat the tacit process knowledge residents hold about local drainage as anecdote rather than data; none operationalizes it as a quantitative calibration input to a probabilistic damage model. Bayesian prior elicitation provides a principled mechanism for translating stakeholder judgment into probability distributions [27, 28], but has not been applied to community flood knowledge. Whether community knowledge and GIS data function as substitutes, complements, or redundant sources remains untested.

2.4 Flood decision support for resource-constrained municipalities: Federal frameworks address adjacent pieces in isolation: the FEMA benefit-cost toolkit evaluates individual projects [29]; the NIST Community Resilience Planning Guide supports capital planning [30, 31]; hazard mitigation programs govern drainage and structural investments without linking them to building damage probabilities [32]; and FEMA retrofit guidance addresses building-by-building measures without a community-scale framework [33, 34]. None couples drainage investment to building damage probabilities through a physical model of the building stock, so nothing carries a completed retrofit back into future flood behavior, and none allows drainage and building hardening to be compared on the same probabilistic basis. None is designed for small municipalities; the hazard mitigation planning literature documents that plan quality falls systematically with municipal size [1, 2, 35]. The fragility gap propagates directly: none addresses pre-code masonry or duration-dependent damage states, none provides building-level predictions from forecast precipitation for pre-storm staging, and no framework improves as a community accumulates events.

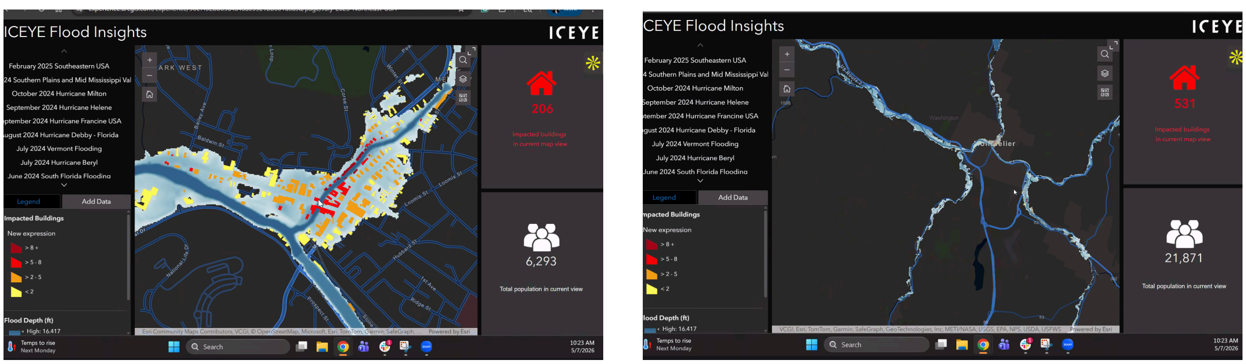
2.5 Gap summary: The pieces exist; the loop does not. Mass balance models are uncharacterized at building-level duration, fragility frameworks cover neither pre-code masonry nor duration, community knowledge has never been a quantitative calibration input, and no decision framework connects drainage investment to building damage probabilities. NIST SP 1310 [11] calls for analogous frameworks for flood as future work [11, Sec. 6.1.2]; no existing tool answers that call for this building stock. The research plan closes the loop.

3 Partner site and preliminary results: The CPS is developed, calibrated, and validated in Montpelier, Vermont across two flood events (2023 and 2024); PI Napolitano has been working directly with the Montpelier community since 2023 [36]. Montpelier, Vermont's capital, sustained catastrophic flooding in July 2023 (80% of its National Register Historic District damaged) and again in 2024, one of over 45 such events since 1900 at the Winooski/North Branch confluence. Its downtown core of 19th-century load-bearing

Figure 1: Montpelier during the July 2023 flood.

masonry, largely pre-dating the 1933 ANSI code, is this proposal's pre-code archetype, with moderate institutional capacity (emergency manager, SHPO support, planning department) but no hydrodynamic simulation infrastructure. The framework transfers to any municipality meeting four conditions, specifically riverine flooding with building-level standing water persisting roughly five or more days; pre-1940 construction as a meaningful share of the flood-exposed inventory; no sustained in-house hydraulic modeling capacity; and USGS gauge coverage on the primary watershed. The first condition is defined on standing-water persistence in the building stock, not on the shorter river-stage exceedance (Preliminary Results below). It is this slow local drainage, not the river crest, that drives duration-governed masonry degradation and that the mass balance model targets.

3.1 Preliminary results: SAR building-level data, Montpelier 2023 and 2024: SAR imagery captures flood extent through cloud cover and at night, when optical imagery fails. Open Sentinel-1 (C-band, 10 m resolution) and commercial ICEYE (X-band, <1 m resolution) both cover the two Vermont events at the meter-scale, few-hour cadence this framework requires (Subtask 1.1). In particular, ICEYE commercial SAR provides building-level flood depth and inundation maps fusing SAR amplitude imagery with LIDAR and water gauge measurements, at revisit cadences as fine as six hours during dense acquisition sequences, across the inundation and recovery period of both Montpelier flood events. Through an NDA data access agreement, ICEYE data for 271 downtown pre-code masonry buildings, each with ICEYE-measured duration, peak depth, and field-recorded damage state, will be analyzed for both the 2023 and 2024 events (Subtask 1.1). The 271-building dataset constitutes the empirical foundation for fragility surface calibration (2023) and held-out validation (2024); the 2024 data is not used in any model fitting (Fig. 2).



(a) July 2023 (calibration event) (b) July 2024 (held-out validation event)

Figure 2: **CHECK WITH CHRISTELLE WHETHER THESE FIGURES ARE OK TO USE OR IF ICEYE PREFERS DIFFERENT ONES.** ICEYE building-level flood depth, Montpelier downtown core, for both study events. Building footprints are colored by peak inundation depth (yellow: <2 ft; orange: 2–5 ft; red: 5–8 ft; dark red: >8 ft). Duration brackets, as fine as six hours between successive overpasses, were extracted separately for all 271 buildings across both events (Subtask 1.1).

3.2 Preliminary results: field damage assessment, Montpelier 2023: PI Napolitano's team conducted a field damage assessment of the 271 downtown masonry buildings before the July 2024 event, so the recorded states reflect 2023 damage only and the 2024 event remains fully held out. Damage states were assigned from combined field inspection, city records, and owner reports using three ordered states: no damage, cosmetic damage (e.g. water staining, minor plaster loss), and structural damage (e.g. mortar deterioration, masonry cracking, foundation compromise). The majority of buildings sustained damage, consistent with the event severity (80% of the National Register Historic District affected). The three-state ordinal classification provides the resolution needed for duration-dependent fragility surface calibration in Subtask 1.4. Each of the 271 buildings in the ICEYE dataset (Fig. 2) carries a field-assigned damage state on this three-level scale, paired building by building with ICEYE-measured depth and duration. Approximately 75 sustained no damage, 120 cosmetic damage, and 76 structural damage.

3.3 Preliminary results: Winooski watershed drainage timescale: Recession analysis of USGS gauge 04286000 (Winooski River at Montpelier; 397 mi^2) yields watershed-scale timescales of $\tau = 37\,\text{h}$ ($R^2 = 0.91$, 2023) and 45 h ($R^2 = 0.96$, 2024) from 15-minute discharge records (Fig. 3). Three observations anchor the research plan. First, the timescale is stable at the day-to-multi-day order masonry degradation operates on and recoverable from the gauge alone; Subtask 1.2 disaggregates it to sub-basin scale from national GIS layers. Second, the 2023 recession departs from a single exponential beyond roughly 60 hours, consistent with standing water in low-lying sub-basins outlasting the river recession, motivating sub-basin $\hat{\tau}_k$ estimates over one watershed constant. Third, the events differ nearly twofold in peak magnitude and the 2024 peak stage stayed below minor flood stage even as downtown parcels took water, so the design tests transfer across event magnitudes. The standing-water distinction is concrete: the 2023 gauge stood above NWS minor flood stage for 28 hours, yet building interiors were pumped for four to six days [37–39]. Building-level standing water, T_i, thus outlasted river-stage exceedance by days, the separation sub-basin timescales $\hat{\tau}_k$ are designed to capture. A prototype stage-to-DEM projection (1 m 3DEP terrain, gauge datum 499.87 ft NAVD88) of the 2023 crest reproduces river-corridor inundation but underpredicts the flooded core by 0.8–1.5 m, implying a water-surface slope of 0.6–1.0 m/km, the parameter the ICEYE depth field calibrates (Subtask 1.2).

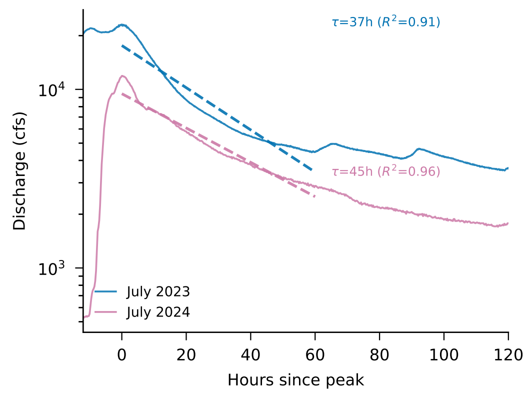


Figure 3: Recession analysis, USGS 04286000 (Winooski at Montpelier), 15-min discharge. Dashed: exponential fits over 60 h windows.

4 Research plan: RO1 (Napolitano, Wauthier, Busse) addresses RQ1 through five subtasks (1.1–1.5); RO2 (Napolitano with all PIs) addresses RQ2 through five (2.1–2.5). Together the objectives close the loop at three timescales: hours before a peak (NOAA forecasts drive staging via Plan/Forecast mode), after each event (community priors update from what the flood revealed, Update Model mode), and over years (retrofit decisions feed back through $\hat{\tau}_k$ and fragility updates, changing the system's next prediction). Human decisions are the actuators at every timescale, making the community a functional component of the control loop. The research questions emerged from PI Napolitano's post-2023 Montpelier engagement, where community partners asked where limited mitigation dollars belong.

Research Objective 1 (RO1, leads: Napolitano, Wauthier, and Busse): Develop and validate a closed-loop CPS coupling a GIS-parameterizable mass balance model with community knowledge in a Bayesian framework and duration-dependent fragility surfaces for pre-code masonry archetypes, and establish whether the system is decision-adequate for retrofit allocation and pre-storm resource staging without hydraulic simulation, calibrated on Montpelier 2023 and validated on the held-out 2024 event.

RO1 builds the first three capabilities identified in the state of the art. Subtask 1.1 processes ICEYE SAR imagery (with open Sentinel-1 for flood-extent context) into the interval-censored duration brackets and peak depths that calibrate the framework. Subtask 1.2 develops the GIS-parameterizable mass balance model that estimates how long water stays. Subtask 1.3 fuses mass balance predictions with community knowledge priors in a Bayesian framework, where community knowledge enters as the building-level adjustment δ_i^{comm} supplied and characterized by RO2.

Subtask 1.4 translates the resulting duration and depth estimates into damage through duration-dependent fragility surfaces for pre-code masonry. ICEYE provides one-time calibration data for the error model and fragility surface; the operational system runs entirely on USGS gauge data and NOAA precipitation, which are free and nationally available. Subtask 1.5 validates the full RO1 system against the 2024 held-out event and characterizes decision-adequacy for scenario-based retrofit ranking and forecast-driven staging.

Subtask 1.1: SAR Data Processing, Interpretation, and Likelihood Characterization (lead: Wauthier). Subtasks 1.1 and 1.2 run in parallel in Year 1 and are independent until both feed Subtask 1.3. PI Wauthier processes ICEYE commercial SAR data for the 271 downtown Montpelier buildings across both events, extracting building-level peak depth and per-overpass wet/dry status. Each building's duration is delivered as a bracket between last-wet and first-dry overpasses (as fine as six hours where acquisitions are dense), carried through the framework as an explicit interval. Outputs are per-building brackets and peak depths for both events with per-scene classification error rates; comparing mass balance predictions against the 2023 brackets calibrates σ^2_{MB} in Subtask 1.3. Cadence simulation shows scenes spread across the drainage window carry more calibration information than scenes clustered near the peak. Open Sentinel-1 is processed for flood-extent context; at $\sim$10 m resolution it cross-checks ICEYE classifications rather than feeding the building-level likelihood.

CHRISTELLE: ADD ICEYE PROCESSING PIPELINE DESCRIPTION AND DEPTH/DURATION MEASUREMENT FLAGS. CONFIRM SCENE CADENCE FOR BOTH EVENTS.

Subtask 1.2: GIS-Parameterizable Mass Balance and Depth Projection (lead: Busse). Subtask 1.2 develops two complementary models, both parameterized entirely from national GIS layers, a mass balance model that estimates drainage timescale $\hat{\tau}_k$ per sub-basin, and a stage-to-DEM projection that estimates building-level peak depth d_i. PI Busse develops the mass balance model for the Winooski watershed above Montpelier, estimating $\hat{\tau}_k$ for each drainage sub-basin from watershed delineation and contributing area from USGS 3DEP and NHD StreamStats, effective impervious fraction from NLCD, soil infiltration rates from SSURGO, and real-time discharge from USGS station 04286000. Each sub-basin is modeled as a linear reservoir, the storage-discharge form whose signature is the exponential recession observed in the preliminary gauge analysis:

$$\frac{dS_k}{dt} = I_k(t) - Q_k, \qquad Q_k = C_k \frac{S_k}{S_k^{\max}} = \frac{S_k}{\hat{\tau}_k}, \qquad \hat{\tau}_k = \frac{S_k^{\max}}{C_k} \tag{1}$$

where S_k is stored floodwater volume in sub-basin k, $I_k(t)$ is inflow from precipitation and upstream routing, $S_k^{\max}$ is the storage capacity of the sub-basin's low-lying terrain from 3DEP, and C_k is its effective conveyance and infiltration capacity from NHD channel properties, NLCD imperviousness, and SSURGO infiltration rates. Outflow is closed as a linear function of fractional storage, $Q_k = C_k\, S_k/S_k^{\max}$, so once inflow subsides the storage decays exponentially and $\hat{\tau}_k = S_k^{\max}/C_k$ is the genuine e-folding recession constant, consistent with the exponential falling limbs fit in the preliminary gauge analysis ($\hat{\tau} = 37$ and 45 h). The drainage timescale $\hat{\tau}_k$ represents the expected duration of standing water under gravitational drainage and serves as the physically grounded prior mean on inundation duration; the model is designed so that any municipality with a USGS gauge ID and access to standard national GIS layers can parameterize it without field data collection. Sub-basins are delineated from 3DEP flow paths at the scale where storage and conveyance properties are resolvable from national layers, on the order of ten to twenty across the flood-exposed core; terrain-partition sensitivity analysis confirms this sits inside the plateau where finer partitioning adds degrees of freedom without adding signal. Building counts per sub-basin are strongly skewed; the model is deliberately lumped, with per-building variation carried by δ_i^{comm} and the error variance. Inundation duration at building i in sub-basin k is modeled as:

$$T_i \sim \mathrm{LogNormal}(\mu_i,\ \sigma^2_{\mathrm{MB}}) \tag{2}$$

where the log-scale prior mean is:

$$\mu_i \;=\; \log \hat{\tau}_k \;+\; \delta_i^{\mathrm{comm}} \tag{3}$$

$\hat{\tau}_k$ is the mass balance drainage timescale for sub-basin k and σ^2_{MB} is the building-scale error variance of the GIS-parameterized mass balance. It absorbs both residual local drainage variability and the model's systematic limitations at building scale, so that all model error lives on the prior side of the framework rather than being attributed to the sensing data. σ^2_{MB} is a cross-sectional quantity, the building-to-building spread of durations within a single event, estimated from the 2023 SAR brackets in Subtask 1.3. A recession analysis of fifteen well-fit events in the gauge's 1990–2026 record gives event-to-event variability $\sigma_{\mathrm{event}} = \mathrm{sd}(\ln \hat{\tau}) = 0.30$, physically distinct from the within-event heterogeneity σ_{MB} measures; it serves only as an order-of-magnitude anchor. Sensitivity analysis confirms the calibration is robust to factor-of-two mis-scaling of this anchor ($<$0.01 change in calibrated σ_{MB}), with the 271 brackets carrying over 80% of the identifying weight. Subtask 1.2 delivers the GIS-only prior, setting $\delta_i^{\mathrm{comm}} = 0$ gives the baseline model parameterized entirely from national GIS layers, which constitutes both the operational system for transfer communities and the comparison baseline for RQ2. The building-level adjustment δ_i^{comm}, elicited from community knowledge in Subtask 2.1 (locally known drainage obstructions, historically slow routes, anomalous building conditions), augments this baseline in Subtask 1.3; community knowledge improves the system but is not required to run it. The LogNormal family is appropriate because duration is strictly positive and right-skewed. All duration quantities are conditional on inundation; the error model characterizes duration error, not flood/no-flood classification. Whether a building floods, and to what peak depth, is supplied by a second, deliberately simple computational component, a stage-to-DEM mapping that projects gauge crest stage onto the USGS 3DEP terrain model along a reach-scale water-surface slope:

$$d_i \;=\; \max\big\{\,0,\; (s_g + \beta\, x_i) - z_i\big\} \tag{4}$$

where s_g is the crest water-surface elevation at the gauge, β is the reach-scale water-surface slope, x_i is the along-channel distance of building i upstream of the gauge, and z_i is the building's ground elevation from 3DEP; a building is predicted inundated where $d_i > 0$, so the same mapping supplies both building-level peak depth and the inundation extent. In Plan/Forecast mode the projection takes a scenario crest stage or the NWS forecast. The 2023 ICEYE depth measurements calibrate β and bound building-level error, defining a distribution over d_i rather than a point value; the depth input for the Subtask 1.4 fragility surface is produced by this component. A single reach-scale β is appropriate because the downtown water surface is a smooth, hydraulically connected profile; slope sensitivity analysis bounds the admissible range from documented 2023 flood outcomes. For pre-storm prediction, NOAA forecast precipitation replaces observed discharge; forecast uncertainty propagates through the mass balance to produce wider credible intervals.

MAGGIE: CONFIRM OR REPLACE TAU FORMULATION. PROVIDE SUB-BASIN DISCRETIZATION AND AMC HANDLING (SEE COMMENTS IN SOURCE).

Subtask 1.3: Bayesian Fusion Framework (leads: Napolitano, Wauthier, and Busse). Subtask 1.3 formulates building-level inundation duration as a state-estimation problem, recovering a latent physical quantity from two information sources with two distinct error structures. The mass balance model, whose building-scale prediction error is carried by the prior variance σ^2_{MB}, and the SAR observations, whose uncertainty is a sampling property of the acquisition schedule rather than an instrument noise floor. A building's inundation duration is never observed directly. Each SAR overpass records only whether the building is inundated at that instant, so the observations constrain the true duration to a bracket $T_i \in [L_i, U_i]$, where L_i is the elapsed time to the last overpass at which building i is observed inundated and U_i the elapsed time to the first overpass at which it is observed dry; the bracket width is the revisit spacing of the acquired scenes. A building still observed wet at the final acquired scene is right-censored, with U_i unbounded; the same survival machinery applies, with likelihood $1 - \Phi((\ln L_i - \mu_i)/\sigma_{\mathrm{MB}})$. The likelihood of the observed wet/dry sequence is the prior probability mass on the bracket, $\Phi\left(\frac{\ln U_i-\mu_i}{\sigma_{\mathrm{MB}}}\right)-\Phi\left(\frac{\ln L_i-\mu_i}{\sigma_{\mathrm{MB}}}\right)$, and the posterior is the prior truncated to it:

$$T_i \mid [L_i, U_i] \;\sim\; \text{LogNormal}(\mu_i,\ \sigma_{\mathrm{MB}}^2) \text{ truncated to } [L_i,\ U_i] \tag{5}$$

With dense acquisitions the brackets are hours wide and the posterior sharply localized; with sparse acquisitions the SAR constrain only weakly. Per-scene classification error ϵ_j (Subtask 1.1) generalizes the hard bracket to a product of per-overpass observation probabilities,

$$p(\boldsymbol{y}_i \mid T_i) \;=\; \prod_j (1-\epsilon_j)^{\,\mathbf{1}\{y_{ij}=\mathbf{1}\{t_j<T_i\}\}}\ \epsilon_j^{\,\mathbf{1}\{y_{ij}\neq\mathbf{1}\{t_j<T_i\}\}} \tag{6}$$

where y_{ij} is the wet/dry classification of building i at overpass time t_j; as $\epsilon_j \to 0$ this recovers the hard bracket $T_i \in [L_i, U_i]$, so classification error softens the truncation boundaries without changing the structure. σ_{MB}^2 is calibrated on the 2023 event by maximum likelihood over the 271 interval-censored observations with the GIS-only prior ($\delta_i^{\mathrm{comm}} = 0$). The σ_{event} anchor centers a weak regularizing prior on σ_{MB}; where brackets are wide, the censored likelihood constrains σ_{MB}^2 weakly and the estimate falls back toward the anchor rather than becoming unidentified. Simulation confirms σ_{MB} remains identified from uniform six-hourly coverage down to a single usable post-peak scene, with estimation uncertainty growing continuously rather than collapsing. The fusion that defines the computational core is between the mass balance prior and the community knowledge adjustments δ_i^{comm}; SAR brackets calibrate the error model on 2023 and adjudicate between prior variants on 2024, but the operational system never depends on them. Posteriors are computed independently per building; portfolio-level uncertainty may be marginally underestimated, but priority rankings are robust to this approximation. For operational deployment without SAR, the predictive distribution is $T_i \sim \text{LogNormal}(\mu_i,\ \sigma_{\mathrm{MB}}^2 + P_i)$, where P_i is the tracked variance of the community adjustment from Subtask 2.4, carrying the full building-scale error of the GIS-parameterized system. The 2024 event is the held-out test: the calibrated framework, with all parameters fixed from 2023, is scored against 2024 SAR brackets.

Subtask 1.4: Duration-Dependent Fragility Surface Development (lead: Napolitano). Subtask 1.4 uses the processed ICEYE duration and depth for all 271 buildings (Subtask 1.1), runs in parallel with Subtasks 1.2 and 1.3, and does not depend on the mass balance model. Duration and peak depth are correlated in the Montpelier 2023 data, motivating a two-component approach that separates the duration effect physically before fitting it statistically. The *physical component* sets the functional form of the duration effect from published masonry degradation mechanisms (mortar softening, brick saturation, rising damp), independently of observational confounding. The *empirical component* calibrates the surface parameters from the 2023 field damage observations, using ICEYE-measured depth and interval-censored duration brackets as covariates for all 271 buildings; consistent with Subtask 1.3, duration enters as its bracket, with estimation integrating over each building's bracket by Monte Carlo sampling from the truncated posterior of Equation (5). The fragility surface for damage state ds takes the form:

$$P(DS \geq ds \mid T_i,\ d_i) \;=\; \Phi\left(\frac{\ln T_i - \lambda_{0,ds} - \gamma \ln d_i}{\zeta}\right) \tag{7}$$

where T_i is inundation duration (hours), d_i is peak inundation depth (m), $\lambda_{0,ds}$ is the log-scale median duration capacity for damage state ds at reference depth, γ is the shared depth sensitivity (depth shifts the threshold at which duration governs damage), ζ is the shared log-standard deviation of capacity, and Φ is the standard normal cumulative distribution function. With the three field-observed damage states (no damage, cosmetic, structural), $ds \in \{1, 2\}$ yields two exceedance curves, $P(DS \geq \text{cosmetic})$ and $P(DS \geq \text{structural})$. The thresholds are constrained to be ordered, $\lambda_{0,1} < \lambda_{0,2}$, while γ and ζ are shared across damage states, so the exceedance curves are horizontal translates of one another along the $\ln T_i$ axis and cannot cross at any (T_i, d_i); their successive differences therefore define a valid ordinal damage-state distribution everywhere. Estimation is Bayesian. Published saturation-strength and capillary-absorption curves for historic brick and lime mortar [17–19] supply informative priors, with $\lambda_{0,ds}$ and γ centered at the material-science-derived values and prior variances set from specimen-to-specimen scatter, updated by the interval-censored 2023 field-damage likelihood. The resulting posterior surface is applied unchanged to 2024 for held-out validation. The claim under test is that duration adds explanatory power beyond depth alone. Because duration and depth are collinear in the 271-building sample, this claim is posed not as a naive coefficient t-test on the correlated data but as a comparison of posterior predictive accuracy on held-out 2024 damage between the bivariate model and a depth-only baseline. The Bayesian machinery makes both the identification and the significance claim coherent. The informative priors on $\lambda_{0,ds}$ and γ carry the identification the collinear 2023 covariates cannot, so the posterior shrinks toward the physical anchor where the empirical signal is weak and refines the anchored values in a genuinely data-driven way where it is strong. Duration-depth collinearity is quantified directly from the 2023 covariates. The bivariate versus depth-only comparison uses the interval-censored log predictive score of Equation (8), summed across buildings, on the held-out 2024 event, consistent with the Bayesian framework used throughout. The posterior duration from Subtask 1.3 propagates through Equation (7) by Monte Carlo integration sampling d_i jointly with T_i, so both covariates carry quantified uncertainty. Sensitivity analysis shows the split is cadence-dependent: duration dominates where brackets are wide (depth error ∼4% of uncertainty under 2023 cadence), depth-projection error dominates where dense brackets pin duration tightly. The 271-building sample is predominantly load-bearing brick unreinforced masonry (URM), with a small number of mixed brick-and-stone construction; together these constitute the pre-code archetype the fragility surface targets.

Subtask 1.5: System Validation (leads: all three PIs). Subtask 1.5 validates the full RO1 system against the 2024 held-out event, with all parameters fixed from 2023 calibration. Technical validation scores predictive duration distributions against the 2024 SAR brackets for all 271 buildings on the interval-censored log predictive score, 90% credible-interval consistency, and RMSE against bracket midpoints where cadence makes brackets tight. The 271-building dataset constitutes one draw from the event distribution, so coverage assessment is limited; multi-event calibration assessment is a target for future work. The interval-censored log predictive score for building i is

$$\mathrm{ILS}_i \;=\; -\log\left[\Phi\left(\frac{\ln U_i-\tilde{\mu}_i}{\tilde{\sigma}_i}\right)-\Phi\left(\frac{\ln L_i-\tilde{\mu}_i}{\tilde{\sigma}_i}\right)\right] \tag{8}$$

where $\tilde{\mu}_i$ and $\tilde{\sigma}_i$ are the predictive log-duration parameters for building i with all model components fixed from 2023; it is a proper scoring rule under interval censoring, and the same score adjudicates the prior variants compared in Subtask 2.3. Damage state validation compares fragility predictions against 2024 field damage observations; because 2024 was smaller, the comparison is reported alongside a 2023 cross-validation, and stated relative to the observed damage-state distribution. Decision-adequacy validation assesses whether retrofit priority rankings align with independent rankings from community partners and the APT DRI expert panel (preservation and emergency management professionals, co-led by PI Napolitano); systematic divergence is treated as a finding rather than suppressed. Forecast validation applies the framework retrospectively using NOAA QPF products at 24-, 48-, and 72-hour lead times; prediction accuracy is reported as a function of lead time to characterize actionable forecast horizon.

Key outcomes: RO1 produces three primary outputs: a validated Bayesian framework, with community knowledge entering as a structural calibration input, with characterized decision-adequacy for scenario-based retrofit ranking and forecast-driven staging (Subtasks 1.3 and 1.5); duration-dependent fragility surfaces for the pre-code masonry building stock (Subtask 1.4); and a characterization of when the GIS-parameterized system is decision-adequate, including the forecast lead times over which predictions remain actionable (Subtask 1.5).

Research Objective 2 (RO2, lead: Napolitano with all PIs): Characterize the conditions under which locally held community knowledge, entered as Bayesian calibration priors, improves damage assessment accuracy and calibration relative to GIS-only model runs, and where community knowledge functions as a substitute versus complement to geospatial data; and co-develop with Montpelier community partners a two-mode decision interface and transferable protocol for independent community use.

RO2 addresses RQ2 by characterizing the conditions under which community knowledge, entered as structural calibration input rather than post-hoc validation, improves the CPS outputs developed in RO1, and where it functions as a substitute versus complement to geospatial data.

Subtask 2.1: Community Co-Development and Knowledge Elicitation (lead: Napolitano). PI Napolitano leads annual co-development workshops with emergency managers, historic preservation officials, and building owners in Montpelier across three years. Community partners include Vermont State Historic Preservation Office representatives, the Montpelier planning commission, the city emergency manager, and building owners in the historic downtown core; letters of collaboration from all partners are included in the supplementary materials. Building owners participate as primary knowledge sources. Their observations of which structures held water longest, which basement entries backed up, and which drainage paths ran clear constitute the building-level knowledge GIS cannot capture and that drives the δ_i^{comm} adjustments in Equation (3); community-engaged flood modeling likewise finds that tools succeed when anchored in local experiential knowledge [40, 41].

Year 1 workshops begin with demonstration before elicitation: the team presents a working prototype from GIS-only output and 2023 ICEYE measurements, showing where the model visibly errs relative to partners' experience. This demonstration-first structure earns trust required for meaningful elicitation [40, 42], and Subtask 2.2 interface co-development begins from this baseline.

Year 1 then surfaces locally held knowledge about which buildings held water longest and which drainage routes are historically slow, operationalized as δ_i^{comm} through a two-step protocol. First, partners identify buildings where the GIS-only model visibly errs and characterize direction and magnitude (e.g. "this drain always backs up, so this building holds water roughly twice as long"). Second, relative statements are converted to log-scale offsets: twice as long yields $\delta_i^{\text{comm}} = \log 2 \approx 0.69$; buildings with no specific knowledge receive $\delta_i^{\text{comm}} = 0$. Each adjustment is recorded together with its stated physical mechanism (an obstructed culvert, a below-grade entry, a historically slow drainage route), not only its magnitude; mechanism attribution is what allows Subtask 2.3 to distinguish durable structural knowledge from recall of the error pattern the research team displayed. Adjustment uncertainty is tracked as variance P_i, initialized from elicitation and updated event by event through Subtask 2.4 (Equation (11)). The Year 2 cross-reference against 2023 SAR actuals provides a validity check before the held-out test. Partners also co-develop the Subtask 2.2 decision interface criteria and weighting; priorities are documented at Year 1 close. *Year 2* validates outputs against stated priorities and adds interpretation training for emergency managers. *Year 3* stress-tests independent use and evaluates whether managers can explain outputs without researcher involvement. This structured process constitutes the **CIR** component required by NSF 25-543, in which locally held knowledge enters the CPS as structural calibration input, community partners co-develop the framework architecture, and the process is assessed against documented community-identified priorities throughout.

Subtask 2.2: Two-Mode Decision Interface (lead: Napolitano). The decision interface supports two modes without requiring knowledge of mass balance hydrology or Bayesian inference. It produces two audience tiers [11, 40, 42]: a *technical tier* with probabilistic distributions, credible intervals, ranked retrofit lists, and what-if outputs for emergency managers; and a *plain-language tier* translating the same analysis into one-page building risk summaries and district-level investment comparisons (e.g. "Buildings on Main Street have a 67% probability of damage >$15k in a 2023-like flood; the proposed retention basin reduces that to 43%"). Partners co-design both tiers in Year 1.

In *Plan/Forecast mode* (always available) the user selects a scenario (historical replay, recurrence interval, custom stage, or live NWS forecast); when driven by a live forecast, a staging priority list for pre-positioning barriers is surfaced. In *Update Model mode* (after a flood), community members submit structured observations that update δ_i^{comm}; confirmed retrofits trigger recalculation of $\hat{\tau}_k$. Plan/Forecast mode is prototyped in Year 1; probabilistic loss calculations become operational in Year 2. For each retrofit action a under consideration, the framework computes expected loss as a sum over buildings and damage states, each damage-state probability obtained by integrating the fragility surface of Equation (7) over the calibrated duration posterior of Equation (5) in Montpelier, or the predictive distribution of Subtask 1.3 where no SAR calibration exists:

$$E[\text{loss} \mid a] \;=\; \sum_i \sum_{ds=1}^{D} \big[\, P(DS_i \geq ds \mid a) - P(DS_i \geq ds{+}1 \mid a) \,\big]\, c_{ds}\, L_i \tag{9}$$

where L_i is the replacement value at building i, c_{ds} is the damage-state-specific loss ratio from HAZUS flood damage functions and FEMA benefit-cost damage ratios [9, 29], and the bracketed difference of successive exceedance probabilities is the probability that building i ends in exactly damage state ds, with the highest state D carrying no survivor term. Each $P(DS_i \geq ds \mid a)$ is computed per building by Monte Carlo integration of Equation (7) over building i's duration posterior jointly with its Subtask 1.2 depth distribution under the stage scenario evaluated, draws with no inundation contributing no loss; the outer sum runs over buildings with nonzero inundation probability under that scenario. The action a modifies both the fragility (building hardening raises the capacity threshold $\lambda_{0,ds}$) and the duration posterior (drainage investment reduces $\hat{\tau}_k$ through the mass balance model). Expected loss reduction is one criterion in the multi-criteria decision analysis score:

$$S(a) \;=\; \sum_{j=1}^{J} w_j \cdot V_j(a) \tag{10}$$

where criteria V_j include structural risk reduction (from Equation (9)), retrofit cost, historic preservation value, and community equity; weights w_j are co-developed with community partners in Year 1 workshops through structured pairwise comparison. Each V_j is min-max normalized to $[0, 1]$ before weighting, so w_j encodes trade-off preference rather than raw-unit artifacts.

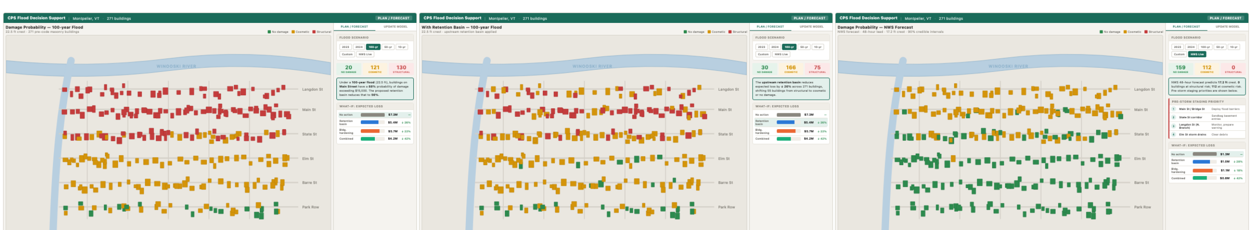


Figure 4: Plan/Forecast mode of the two-mode decision interface (Subtask 2.2), showing building-level damage probabilities for 271 downtown Montpelier buildings. Left: 100-year flood baseline (no action). Center: same scenario with the proposed upstream retention basin, showing reduced structural damage. Right: NWS 48-hour forecast (17.2 ft crest) with pre-storm staging priorities. Each panel includes the what-if expected loss comparison and community priority weights (Equation (10)).

Subtask 2.3: Comparative Assessment of Community Knowledge Contribution (leads: all three PIs). Subtask 2.3 directly addresses RQ2 by running the full CPS under two conditions, with community-informed priors (δ_i^{comm} from Subtask 2.1 workshops) and with GIS-only priors ($\delta_i^{\text{comm}} = 0$), across all 271 buildings for both flood events. Both conditions are scored against SAR brackets on the Subtask 1.5 metrics. Because Year 1 elicitation shows partners the 2023 SAR map, the 2023 comparison is a diagnostic of elicitation mechanics; the held-out 2024 event carries the substantive RQ2 test. Cross-event persistence, evaluated separately for mechanism-attributed and unexplained adjustments, separates structural knowledge from recall. Results are disaggregated by building type, sub-basin, and damage state to characterize where community knowledge and GIS data are substitutes versus complements; bracket width bounds the detectable signal per sub-group. Power analysis confirms 70–80% power at mechanism-attributed rates of 20–30% of the 271 buildings; volume without mechanism quality contributes no power, confirming the Subtask 2.1 attribution requirement is what makes RQ2 testable. Cases where community knowledge underperforms GIS-only are reported in full, since null findings inform the Subtask 2.5 transfer protocol.

Subtask 2.4: Closed-Loop Validation (leads: all three PIs). The loop closes through Bayesian prior updating across events. After each flood, structured observations update δ_i^{comm}; buildings whose observed durations consistently deviate from predictions receive updated priors that carry forward. Formally, δ_i^{comm} follows a random walk with step variance q^2 (non-stationary drainage); each event updates both the estimate and tracked variance P_i,

$$\begin{aligned} \delta_i^{\text{comm},(e+1)} &= \left(1 - w_i^{(e)}\right) \delta_i^{\text{comm},(e)} + w_i^{(e)} \left(\widehat{\ln T_i^{(e)}} - \ln \hat{\tau}_k\right), \qquad w_i^{(e)} = \frac{P_i^{(e)}}{P_i^{(e)} + v_i^{(e)}}, \\ P_i^{(e+1)} &= \left(1 - w_i^{(e)}\right) P_i^{(e)} + q^2 \end{aligned} \tag{11}$$

where $\widehat{\ln T_i^{(e)}}$ and $v_i^{(e)}$ are the posterior mean and variance of log duration from event e, $w_i^{(e)}$ is the Kalman gain, and $P_i^{(e)}$ is the tracked adjustment variance, initialized from Subtask 2.1 elicitation. The variance converges to a steady state set by q; simulation confirms convergence but shows a single event pair constrains q only coarsely, so q is treated as a swept design hyperparameter. After retrofit completion, $\hat{\tau}_k$ is updated from engineering specifications (storage volume and outlet capacity) rather than a full model re-run; building hardening updates $\lambda_{0,ds}$. No drainage retrofits are currently planned; the pathway is demonstrated computationally by simulating a plausible investment and showing the resulting change in rankings. The 2024 event tests whether one observation-update cycle measurably improves accuracy over the 2023-only baseline.

Subtask 2.5: Transferable Protocol (lead: Napolitano with all PIs). The transferable protocol specifies the minimum pathway for a new community. A municipality needs a USGS gauge ID, standard national GIS layer downloads (USGS 3DEP, NLCD, SSURGO, NHD StreamStats), and one co-development workshop to elicit initial δ_i^{comm} values, optionally supported by free Sentinel-1 imagery as a rough visual aid for flood-extent discussion where local recall is difficult, combined with the open-source software package initialized from Montpelier-derived mass balance and fragility parameters. A community operating under this protocol has wider credible intervals than Montpelier because it lacks local ICEYE calibration, and free Sentinel-1 imagery does not close this gap. The expected interval widths, derived from the Subtask 1.3 σ^2_{MB} characterization and the elicitation-initialized $P_i^{(0)}$, are reported in the protocol documentation so communities can assess decision-adequacy before adoption, and intervals narrow with each observation cycle. The protocol's elicitation guidance is informed by the Subtask 2.3 substitute-versus-complement characterization. Where community knowledge adds value, the protocol prioritizes elicitation; where it does not, the protocol directs communities to rely on the GIS-only baseline rather than investing in workshops that produce no measurable improvement.

Key outcomes: RO2 produces four primary outputs: a two-mode interface independently operated by Montpelier partners in Year 3 (Subtask 2.2); a characterization of when community knowledge and geospatial data are substitutes versus complements, including cases where community knowledge adds no value (Subtask 2.3); empirical demonstration that the loop closes, with community observations and retrofit specifications measurably improving posterior accuracy across events (Subtask 2.4); and a transferable protocol enabling any municipality with a USGS gauge to initialize the system from one workshop without a commercial SAR purchase (Subtask 2.5).

5 Broader impacts: The framework gives small municipalities probabilistic damage assessments and pre-storm decision support that does not currently exist, reducing the flood damage burden that falls disproportionately on towns with the fewest resources. Plan/Forecast mode delivers actionable risk before water rises; Update Model mode improves accuracy with each event, without proprietary data or ongoing research support. Operational inputs are only USGS gauge data and NOAA precipitation, both free; the mass balance model is parameterizable from national GIS layers for any municipality with a gauge. A new community needs a gauge ID, standard GIS access, and one workshop; no commercial SAR purchase, no hydrologist, and no hydraulic model is required.

Emergency managers and planning staff gain capacity to interpret probabilistic assessments and run what-if simulations independently. The CIR methodology, documented as a transferable protocol, converts tacit process knowledge into quantitative model inputs measured against a geospatial baseline; resident knowledge becomes measurably load-bearing rather than supplementary narrative. The project supports one postdoctoral researcher (co-advised by PI Napolitano and Co-PI Wauthier, focused on SAR processing and Bayesian fusion) and one PhD student (advised by Co-PI Busse, focused on mass balance modeling and community co-development). Both receive cross-disciplinary training spanning SAR imagery data processing, hydrological mass balance modeling, and probabilistic structural assessment and community-engaged research across all three PIs.

5.1 Potential for transformative impact: Placing drainage investment and building retrofits on the same probabilistic basis resolves an allocation problem that annually re-injures small historic municipalities. The generalizable principle, that free national data fused with tacit process knowledge suffices for the investment decisions these municipalities face, extends to any infrastructure domain with national data and practitioner knowledge but no simulation capacity. The substitutes-versus-complements characterization (RO2) provides the first empirical basis for deciding when elicitation is worth the investment and when national data alone suffices.

5.2 Education, broadening participation, and practitioner upskilling: Two undergraduates per year participate through the BESURE program at Penn State, with recruitment prioritizing underrepresented groups. In Year 3, PI Napolitano leverages her membership in the Heritage Emergency National Task Force (HENTF), connecting all State Historic Preservation Officers nationally, to deliver a webinar series walking SHPOs through initializing the system for their municipalities using the Subtask 2.5 transferable protocol.

5.3 Open science and transferability: The Bayesian fusion code, mass balance parameterization pipeline, and decision interface will be released on GitHub under an MIT license, with documentation enabling any municipality to initialize the system from national GIS data layers. Derived SAR imagery, codes, and building-level inundation products for Montpelier will be shared on DesignSafe-CI; field damage state observations and community workshop documentation will be shared on the Open Science Framework with appropriate anonymization.

6 Project timeline and evaluation: Progress is monitored through monthly PI meetings and annual reviews. RO1 technical development (Subtasks 1.1 to 1.4) runs in Years 1 and 2 parallel to RO2 Year 1 co-development and interface construction; validation (Subtask 1.5) and comparative assessment (Subtask 2.3) follow in Year 2; closed-loop validation (Subtask 2.4), protocol transfer (Subtask 2.5), and Year 3 independent-use stress testing conclude in Year 3. Evaluation milestones are marked in Fig. 5. *Technical evaluation* (M1, end of Year 2) requires building-level duration error at or below $\sim$0.4 log-units RMSE on the 2024 held-out event, 90% credible interval coverage between 0.85 and 0.95, duration-depth bivariate improvement over the depth-only baseline, and actionable 48-hour forecast rankings. *Community engagement evaluation* (M2, end of Year 1) requires structured elicitation covering all 271 buildings with mechanism-attributed adjustments for at least 20% (target 30%); Year 2 community assessment (M3) confirms Year 1 priorities are reflected in framework outputs. *Community knowledge contribution* (M4, end of Subtask 2.3) requires the community-informed model to outperform the GIS-only baseline on at least one disaggregated sub-group, with full results including null findings published. *Adoption* (M5, Year 3) requires emergency managers to operate both interface modes and explain outputs to officials without researcher support; the transferable protocol is released with characterized interval widths for communities without local SAR calibration.

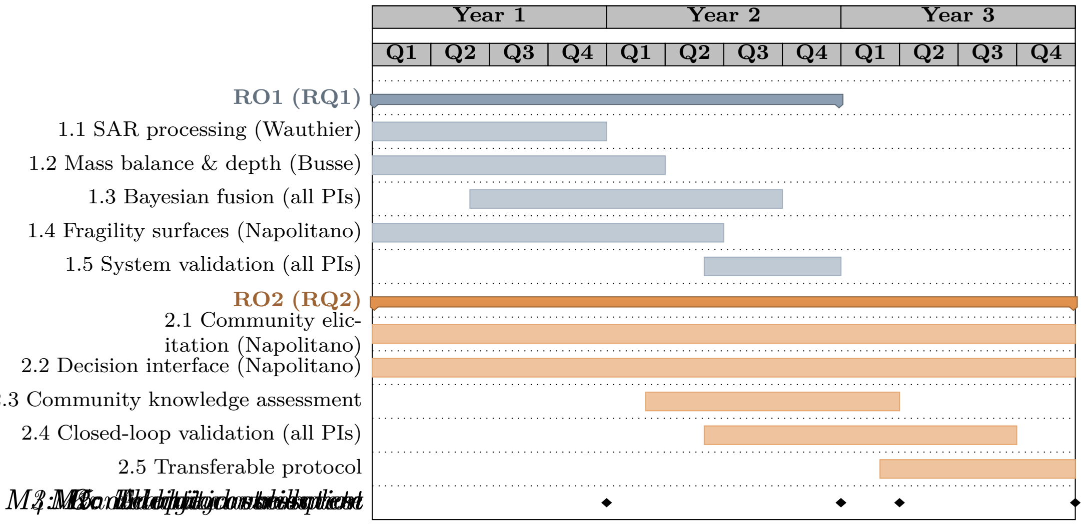


Figure 5: Project timeline. RO1 subtasks (blue) develop the CPS framework in Years 1–2; RO2 subtasks (orange) co-develop with community partners across all three years. Validation (1.5) and comparative assessment (2.3) overlap in Year 2; closed-loop validation (2.4) and transfer (2.5) conclude in Year 3.

7 Results from prior NSF Support: PI Napolitano:

PI Napolitano was PI on NSF CMMI-2222849 ($131,317, 04/2022–02/2025): "RAPID: Examining the Performance of Historic Masonry Structures after the Midwest Tornadoes." Intellectual Merit: This project addressed building code gaps by analyzing historic masonry structures' performance under tornadic loading and assessing retrofitting efficacy, producing the first systematic dataset on tornado damage to unreinforced masonry heritage structures. Broader Impacts: The project enhanced understanding of historic masonry resilience under wind hazards, produced open datasets archived on DesignSafe, and integrated community-centered structural data into engineering education. Publications: [43–48]. Undergraduate research presentations: [49, 50]. Publicly shared datasets: [51–54]. Renewal: N/A.

Co-PI Wauthier: Christelle Wauthier is the PI on NSF EAR-1945417 ($499,999k, 08/01/20-07/31/26): "CAREER: Numerical Modeling of Volcanic Flank Instability Processes". Intellectual Merit: This CAREER proposal addresses why the style, behavior, and timing of the cyclic growth then destruction of volcanic edifices varies across such a wide spectrum. It focuses on flank collapse on ocean island volcanoes and in using modeling to unravel the observed complexity in data sets representing a spectrum of type volcanoes. Broader Impacts: Graduate and undergraduate education, support for underrepresented undergraduate research interns, new educational activities, and FEM and other modeling training for graduate students. Research products: Cooper et al., 2023; Leeburn* et al., 2022; Gonzalez-Santana* et al., 2022, 2024; Gonzalez-Santana* and Wauthier, 2021; Poppe* et al., 2024; Kim* et al., 2024; Wauthier and Ho, 2024; Gonzalez-Santana* et al., 2025, all of which are listed in the References.

Co-PI Busse: Margaret Busse is senior personnel on NSF CBET-2400672 ($1,000,000, 8/2024 - 7/2029): "RAISE: CET: Co-development of engineering energy literacy and engineering identities through campus decarbonization research." Intellectual Merit: This work integrates six component-specific research directions to address fundamental questions associated with net-zero fuels in the context of Combined Heat and Power (CHP) decarbonization. Intellectual Merit of the research lies in the integrated real-world system evaluation approach as well as the evaluation of student energy literacy through their research experience. Broader Impacts: The Broader Impacts include advancement of decarbonization technologies and the development of an energy-literate future engineering workforce. Research Products: None.

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